

```
mysql> use lab4and5and6
Database changed
mysql> show tables;
+-----+
| Tables_in_lab4and5and6 |
+-----+
| customer_address        |
| customer_classification_type |
| customer_contact_info   |
| customer_detail         |
| customer_identification  |
| customer_names          |
| customer_proof_ofid     |
+-----+
7 rows in set (0.02 sec)
```

```
mysql> select * from customer_address;
+-----+-----+-----+-----+-----+-----+
| customer_address_id | customer_address_type | customer_address_value | effective_date | customer_classification_id | customer_identifier |
+-----+-----+-----+-----+-----+-----+
| 1 | HOME | 221B Baker Street | 2024-01-01 | 1 | 1 |
| 4 | HOME | HSR Layout, Bangalore | 2025-01-01 | 1 | 1 |
| 5 | HOME | San Francisco, California | 2026-01-01 | 1 | 1 |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.06 sec)
```

```
mysql> select * from customer_classification_type;
+-----+-----+-----+-----+
| customer_classification_id | customer_classification_type | customer_classification_value | effective_date |
+-----+-----+-----+-----+
| 1 | INDIVIDUAL | RETAIL_CUSTOMER | 2024-01-01 |
| 2 | BUSINESS | CORPORATE_PLUS | 2025-04-01 |
| 3 | VIP | HIGH_NET_WORTH | 2024-01-01 |
+-----+-----+-----+-----+
3 rows in set (0.06 sec)
```

```
mysql> select * from customer_contact_info;
+-----+-----+-----+-----+-----+-----+-----+
| cust_contact_infoid | customer_contact_type | customer_contact_value | effective_date | end_date | start_date | customer_identifier |
+-----+-----+-----+-----+-----+-----+-----+
| 1 | MOBILE | +91-9999999999 | 2024-01-01 | 2030-01-01 | 2024-01-01 | 1 |
| 4 | MOBILE | +91-9988776655 | 2025-01-01 | 2035-01-01 | 2025-01-01 | 1 |
| 5 | MOBILE | +91-9999999999 | 2024-01-01 | 2030-01-01 | 2024-01-01 | 1 |
+-----+-----+-----+-----+-----+-----+-----+
3 rows in set (0.03 sec)
```

```
mysql> select * from customer_detail;
+-----+-----+-----+-----+-----+-----+-----+-----+
| customer_identifier | customer_type | customer_full_name | customer_gender | customer_preferred_language | customer_status | customer_country_of_origin | customer_classification_id |
+-----+-----+-----+-----+-----+-----+-----+-----+
| 1 | 1994-06-21 | Ravi Kumar | MALE | KN | ACTIVE | INDIA | 1 |
+-----+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.02 sec)
```

```
mysql> select * from customer_identification;
+-----+-----+-----+-----+-----+
| customer_identification_id | customer_identification_item | customer_identification_type | effective_date | customer_identifier |
+-----+-----+-----+-----+-----+
| 1 | 9999-8888-7777 | Passport | 2026-05-01 | 1 |
+-----+-----+-----+-----+-----+
1 row in set (0.02 sec)
```

```
mysql> select * from customer_names;
+-----+-----+-----+-----+-----+-----+
| customer_name_id | customer_name_type | customer_name_value | effective_date | customer_classification_id | customer_identifier |
+-----+-----+-----+-----+-----+-----+
| 1 | LEGAL | Rohit Verma | 2024-01-01 | 1 | 1 |
| 2 | DISPLAY | Rohit | 2024-01-01 | 1 | 1 |
| 3 | LEGAL | Ravi Kumar | 2025-01-01 | 1 | 1 |
| 4 | DISPLAY | Rish | 2026-02-01 | 1 | 1 |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.03 sec)
```

```
mysql> select * from customer_proof_ofid;
+-----+-----+-----+-----+-----+-----+-----+
| customer_proofofid_id | effective_date | end_date | cust_proof_of_id_type | cust_proof_of_id_value | start_date | customer_identifier |
+-----+-----+-----+-----+-----+-----+-----+
| 1 | 2024-01-01 | 2034-01-01 | PAN | ABCDE1234F | 2024-01-01 | 1 |
| 2 | 2025-01-01 | 2035-01-01 | PAN | ABCDE4321K | 2025-01-01 | 1 |
| 3 | 2024-01-01 | 2034-01-01 | EPIC | ABCDE1234E | 2024-01-01 | 1 |
+-----+-----+-----+-----+-----+-----+-----+
3 rows in set (0.03 sec)
```

✓ Lab4and5and6 [boot]

✓ src/main/java

✓ com.example.demo

>  Lab4and5and6Application.java

✓ com.example.demo.controllers

>  CustomerAddressController.java

>  CustomerClassificationController.java

>  CustomerContactInfoController.java

>  CustomerDetailController.java

>  CustomerIdentificationController.java

>  CustomerNameController.java

>  CustomerProofOfIdController.java

✓ com.example.demo.DTO

>  custAddressDTO.java

>  custClassifDTO.java

>  custContactInfoDTO.java

>  custDetailDTO.java

>  custIdentifDTO.java

>  custNameDTO.java

>  custProofOfIdDTO.java

✓ com.example.demo.entity

>  CustomerAddress.java

>  CustomerClassificationType.java

>  CustomerContactInformation.java

>  CustomerDetail.java

>  CustomerIdentification.java

>  CustomerName.java

>  CustomerProofofId.java

✓ com.example.demo.mapping

>  CustDetailMapper.java

✓ com.example.demo.repository

>  CustomerClassificationRepo.java

>  CustomerContactInformationRepo.java

>  CustomerDetailRepo.java

>  CustomerIdentificationRepo.java

>  CustomerNameRepo.java

>  CustomerProofofIdRepo.java

>  CustomerAddressRepo.java

✓ com.example.demo.service

>  CustomerAddressService.java

>  CustomerClassificationService.java

>  CustomerContactInfoService.java

>  CustomerDetailsService.java

>  CustomerIdentificationService.java

>  CustomerNameService.java

✓ com.example.demo.service

>  CustomerAddressService.java

>  CustomerClassificationService.java

>  CustomerContactInfoService.java

>  CustomerDetailsService.java

>  CustomerIdentificationService.java

>  CustomerNameService.java

>  CustomerProofOfIdService.java

✓ com.example.demo.serviceImpl

>  custDetailsServiceImpl.java

>  CustomerAddressServiceImpl.java

>  CustomerClassificationServiceImpl.java

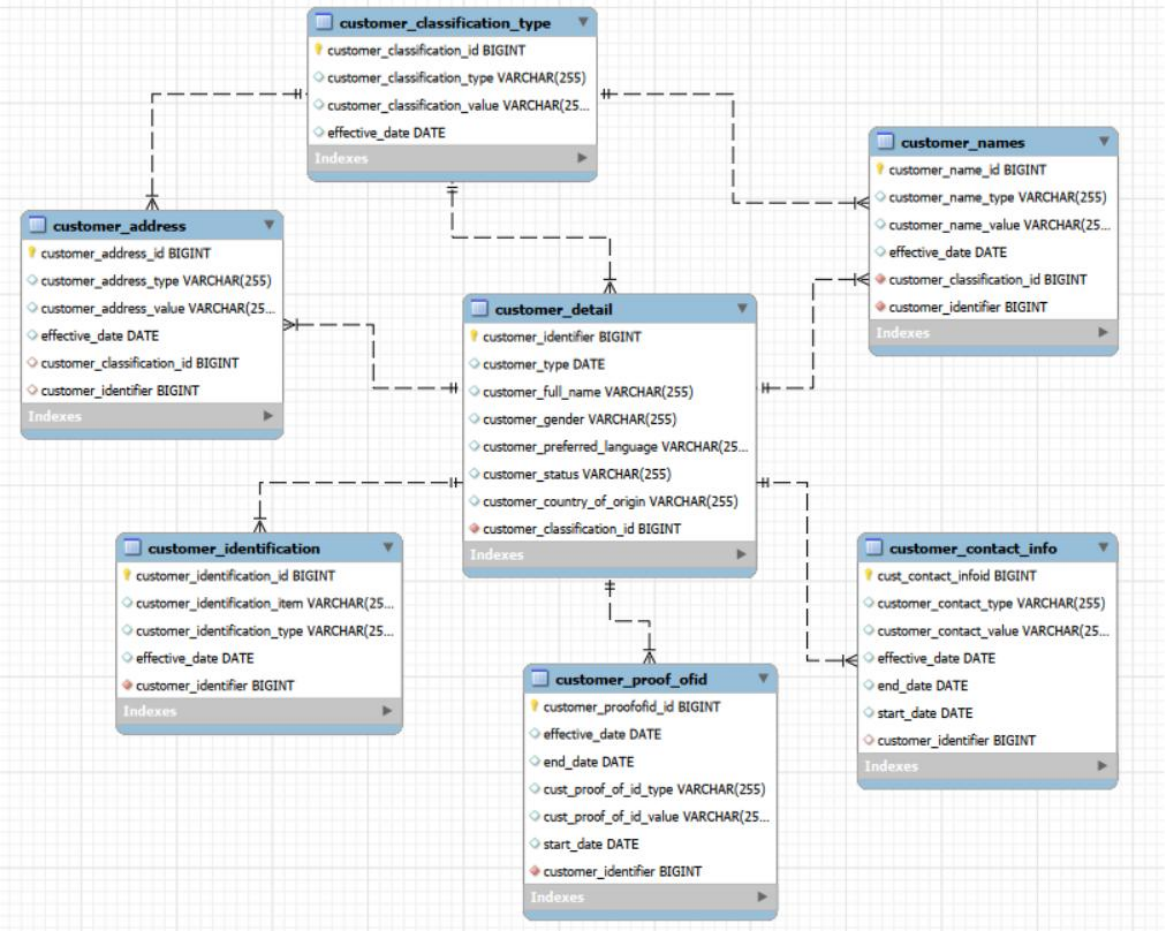
>  CustomerContactInfoServiceImpl.java

>  CustomerIdentificationServiceImpl.java

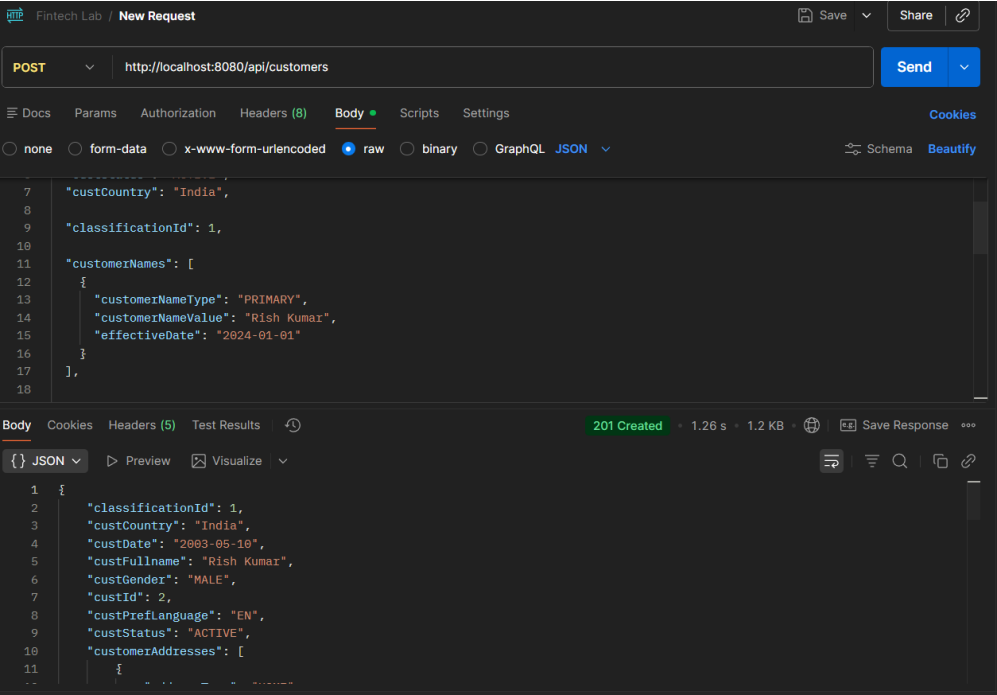
>  CustomerNameServiceImpl.java

>  CustomerProofOfIdServiceImpl.java

On Reverse engineering the database:



sending a post request to the customer detail



customer detail record created in the table

```
mysql> select * from customer_detail;
```

customer_identifier	customer_type	customer_full_name	customer_gender	customer_preferred_language	customer_status	customer_country_of_origin	customer_classification_id
1	1994-06-21	Ravi Kumar	MALE	KN	ACTIVE	INDIA	1
2	2003-05-10	Rish Kumar	MALE	EN	ACTIVE	India	1

2 rows in set (0.06 sec)

Similarly a record is created in each and every class

```
mysql> select * from customer_identification;
```

customer_identification_id	customer_identification_item	customer_identification_type	effective_date	customer_identifier
1	9999-8888-7777	Passport	2026-05-01	1
4	1234-5678-9012	AADHAAR	2024-01-01	2

2 rows in set (0.01 sec)

```
mysql> select * from customer_names;
```

customer_name_id	customer_name_type	customer_name_value	effective_date	customer_classification_id	customer_identifier
1	LEGAL	Rohit Verma	2024-01-01	1	1
2	DISPLAY	Rohit	2024-01-01	1	1
3	LEGAL	Ravi Kumar	2025-01-01	1	1
4	DISPLAY	Rish	2026-02-01	1	1
5	PRIMARY	Rish Kumar	2024-01-01	1	2

5 rows in set (0.01 sec)

```
mysql> select * from customer_proof_ofid;
```

customer_proofofid_id	effective_date	end_date	cust_proof_of_id_type	cust_proof_of_id_value	start_date	customer_identifier
1	2024-01-01	2034-01-01	PAN	ABCDE1234F	2024-01-01	1
2	2025-01-01	2035-01-01	PAN	ABCDE4321W	2025-01-01	1
3	2024-01-01	2034-01-01	EPIC	ABCDE1234E	2024-01-01	1
4	2024-01-01	NULL	AADHAAR	1234-5678-9012	2024-01-01	2

4 rows in set (0.01 sec)

```
mysql> select * from customer_contact_info;
```

cust_contact_infoid	customer_contact_type	customer_contact_value	effective_date	end_date	start_date	customer_identifier
1	MOBILE	+91-9999999999	2024-01-01	2030-01-01	2024-01-01	1
4	MOBILE	+91-9988776655	2025-01-01	2035-01-01	2025-01-01	1
5	MOBILE	+91-9999999999	2024-01-01	2030-01-01	2024-01-01	1
6	EMAIL	rish@gmail.com	2024-01-01	NULL	2024-01-01	2

4 rows in set (0.00 sec)

Customer classification table remains the same as we did not add any new classification type

```
mysql> select * from customer_classification_type;
```

customer_classification_id	customer_classification_type	customer_classification_value	effective_date
1	INDIVIDUAL	RETAIL_CUSTOMER	2024-01-01
2	BUSINESS	CORPORATE_PLUS	2025-04-01
3	VIP	HIGH_NET_WORTH	2024-01-01

3 rows in set (0.00 sec)

Sending a PUT request and updating the customer proof of ID:

PUT ▼ http://localhost:8080/api/proof/4

Docs Params Authorization Headers (8) **Body** • Scripts Settings

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL **JSON** ▼

```
1 {
2   "proofofIdType": "US PASSPORT",
3   "proofofIdValue": "M1234567",
4   "effectivDate": "2024-02-01",
5   "startDate": "2024-02-01",
6   "endDate": "2028-02-01"
7 }
8
```

Body Cookies Headers (5) Test Results ↻ **200 OK** • 39 ms • 333 B • 🌐 📄

{} JSON ▼ ▶ Preview 🖼 Visualize ▼

```
1 {
2   "customerProofId": 4,
3   "proofofIdType": "US PASSPORT",
4   "proofofIdValue": "M1234567",
5   "effectivDate": "2024-02-01",
6   "startDate": "2024-02-01",
7   "endDate": "2028-02-01",
8   "custId": null
9 }
```

Updated:

```
mysql> select * from customer_proof_ofid;
```

customer_proofofid_id	effective_date	end_date	cust_proof_of_id_type	cust_proof_of_id_value	start_date	customer_identifier
1	2024-01-01	2034-01-01	PAN	ABCDE1234F	2024-01-01	1
2	2025-01-01	2035-01-01	PAN	ABCDE4321K	2025-01-01	1
3	2024-01-01	2034-01-01	EPIC	ABCDE1234E	2024-01-01	1
4	2024-02-01	2028-02-01	US PASSPORT	M1234567	2024-02-01	2

4 rows in set (0.00 sec)

Using get request to get all the customer names:

GET http://localhost:8080/api/customernames

Send

Docs Params Authorization Headers (8) Body Scripts Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify

Body Cookies Headers (5) Test Results 200 OK • 189 ms • 851 B Save Response

{ } JSON Preview Visualize

```
1 [
2   {
3     "custNameId": 1,
4     "customerNameType": "LEGAL",
5     "customerNameValue": "Rohit Verma",
6     "effectiveDate": "2024-01-01",
7     "classificationId": 1,
8     "custId": 1
9   },
10  {
11    "custNameId": 2,
12    "customerNameType": "DISPLAY",
13    "customerNameValue": "Rohit",
14    "effectiveDate": "2024-01-01",
15    "classificationId": 1,
16    "custId": 1
17  },
18  {
19    "custNameId": 3,
20    "customerNameType": "LEGAL",
21    "customerNameValue": "Ravi Kumar",
22    "effectiveDate": "2025-01-01",
23    "classificationId": 1,
```

Similarly the above operations are valid and working for all the other entities/tables also